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from sklearn.impute import SimpleImputer

imputer = SimpleImputer(
    missing_values=np.nan,      # What to consider as missing (default: np.nan)
    strategy='mean',           # Strategy: 'mean', 'median', 'most_frequent', 'constant'
    fill_value=None,           # Used when strategy='constant'
    copy=True,                 # Whether to make a copy of the data
    add_indicator=False,       # Whether to add missing indicator
    keep_empty_features=False  # Whether to keep features with only missing values
)

df_imputed = imputer.fit_transform(df)

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import pandas as pd
import numpy as np
from sklearn.impute import SimpleImputer

# Create sample data
data = {
    'Age': [25, 30, np.nan, 35, 40, np.nan, 28],
    'Salary': [50000, np.nan, 70000, 80000, np.nan, 60000, 55000],
    'Experience': [2, 5, np.nan, 10, 3, 7, np.nan],
    'Department': ['IT', 'HR', 'IT', np.nan, 'Finance', 'IT', 'HR']
}

df = pd.DataFrame(data)

print("BEFORE IMPUTATION:")
print(df)
print("\nMissing values:")
print(df.isna().sum())

# Impute numerical columns
numerical_imputer = SimpleImputer(strategy='mean')
numerical_cols = ['Age', 'Salary', 'Experience']
df[numerical_cols] = numerical_imputer.fit_transform(df[numerical_cols])

# Impute categorical columns - CORRECTED
categorical_imputer = SimpleImputer(strategy='most_frequent')
department_imputed = categorical_imputer.fit_transform(df[['Department']])
df['Department'] = department_imputed.ravel() # Convert 2D to 1D # [['IT'], ['HR'], ['IT'], ['IT'], ['Finance'], ['IT'], ['HR']]

print("\nAFTER IMPUTATION:")
print(df)
print("\nMissing values:")
print(df.isna().sum())

```

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BEFORE IMPUTATION:
   Age  Salary  Experience  Department
0  25.0  50000.0         2.0         IT
1  30.0     NaN         5.0         HR
2   NaN  70000.0         NaN         IT
3  35.0  80000.0        10.0         NaN
4  40.0     NaN         3.0     Finance
5   NaN  60000.0         7.0         IT
6  28.0  55000.0         NaN         HR

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Missing values:
Age          2
Salary       2
Experience    2
Department    1
dtype: int64

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AFTER IMPUTATION:
   Age  Salary  Experience  Department
0  25.0  50000.0         2.0         IT
1  30.0  63000.0         5.0         HR
2  31.6  70000.0         5.4         IT
3  35.0  80000.0        10.0         IT
4  40.0  63000.0         3.0     Finance
5  31.6  60000.0         7.0         IT
6  28.0  55000.0         5.4         HR

```

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Missing values:
Age          0
Salary       0

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Experience    0  
Department    0  
dtype: int64
```